

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS (NC State, May 2026). Built RAG pipelines, LLM eval harnesses, and agentic AI flows in production: per-tenant RAG isolation for fintech/healthcare, vendor-neutral 3-model benchmark (Claude/GPT-4o/Gemini) with hallucination rate and latency tracking, and event-driven orchestration at 2,000 async req/sec via Temporal.io. Multi-model LLM pipelines across 3 providers with MCP tooling across 7 production deployments at an AI startup. Ready to own the AI backend at a dev tools company.

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern

- Built **multi-model AI agent tooling across 3 providers and 8+ models** (Anthropic Claude, OpenAI GPT-4o, Google Gemini) using MCP and prompt engineering; **7 production deployments**, 394 commits across TypeScript, Node.js, PostgreSQL, Redis, Temporal.io.
- Implemented document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage), real-time event pipelines (Pusher), and RBAC system -- all in production serving real users.

Mphasis (HP Company) Jun -- Aug 2024

Software Developer Intern

- Built AI-driven claim automation workflow (OCR, NLP, ML) improving accuracy 25%; automated AWS/GCP deployments with Grafana observability and red-teaming support.

Capgemini Jun -- Aug 2023

AI/ML Intern

- Built Hadoop + Spark + Kafka distributed pipelines for support ticket analysis; cut processing time 15% and latency 20%.

PROJECTS

LLM Eval Harness 2025

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro on 2,000-example frozen test set (grounded QA, multi-hop QA, FEVER claim verification). Tracked ExactMatch, grounded **hallucination rate**, p50/p95 latency, cost/call; statistical significance via paired bootstrap CIs and McNemar test. Streamlit dashboard for per-model failure analysis.

Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker

Glance AI -- Privacy-First RAG System 2025

Compliance-ready **RAG pipeline** for fintech/healthcare with **per-tenant data isolation at the vector retrieval layer**, preventing context leakage across tenants. Deterministic output architecture for regulated data environments.

Python, LLMs, Vector Databases, RAG

VectorLift -- Semantic Search Engine 2026

Production-grade semantic search: BM25 baseline, dense bi-encoder (sentence-transformers), cross-encoder re-ranker. **Kafka-driven data ingestion pipeline** into Elasticsearch and Qdrant vector store. Evaluated 6 pipeline configs with NDCG@10, MRR@10, MAP; Prometheus/Grafana observability.

Python, PyTorch, Elasticsearch, Qdrant, Kafka, FastAPI, Prometheus, Grafana, Docker

EDUCATION

MS Computer Science -- North Carolina State University Aug 2024 -- May 2026

AI, Operating Systems, Data Science, Algorithms, Software Engineering

B.Tech Computer Science -- Mumbai University Aug 2020 -- May 2024

CGPA 9.32/10 | Machine Learning, Big Data Infra, Data Warehousing, Algorithms

SKILLS

AI/ML: RAG pipelines, LLM evals, prompt engineering, Claude/OpenAI/Gemini APIs, MCP, Hugging Face, PyTorch, Qdrant, Elasticsearch

Languages: Python, TypeScript, JavaScript, Java, SQL, C

Infra: Temporal.io, Kafka, Docker, AWS, GCP, PostgreSQL, Redis, Grafana, Prometheus, Node.js, React

Awards: HackNC24 Top 10 (100+ teams), 6+ national hackathon wins, IEEE Published (Mapify), Top 1% Sanjeevani (1100+ teams)